

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(5,400,000)	(5,142,857)	5,000,000	4,761,905
2	5.113%	5.226%	(5,400,000)	(4,887,451)	5,225,748	4,729,739
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
						(10,030,308)
						9,491,644
						Receiver 561,288
			Fixed rate	5.400%		
			Notional	100,000,000	NPV	22,623

Swaption model		1	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.0062
2	5.110%	5.113%	1x2	0.9051	d_2	-0.2562
3	5.300%	5.312%	2x3			
4	5.470%	5.494%	3x4			
5	5.600%	5.635%	4x5			
F	5.226%		$A =$	0.9051		
σ	25.0%				$XN(-d_2) - FN(-d_1)$	0.006202
T	1		Notional	100,000,000	Receiver swaption	561,288
X	5.400%				(in bps)	56.1

Figure 6.10: The Value of a 2nc1 with a Fixed Coupon of 5.40% when $\sigma = 25\%$

6.3 Index Amortizing Swaps

Index amortizing swaps were extremely popular in the 1990s but are practically defunct now. An elegant trade from a more civilized age, this trade merits discussion given its historical significance. It demonstrated both the

SWAPS WITH EMBEDDED OPTIONS

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(5,417,254)	(5,159,289)	5,000,000	4,761,905
2	5.113%	5.226%	(5,417,254)	(4,903,067)	5,225,748	4,729,739
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
				(10,062,357)		9,491,644
					Receiver	570,713
Fixed rate			5.417%			
Notional			100,000,000	NPV		
				(0)		

Swaption model		1	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.0190
2	5.110%	5.113%	1x2	0.9051	d_2	-0.2690
3	5.300%	5.312%	2x3			
4	5.470%	5.494%	3x4			
5	5.600%	5.635%	4x5			
F	5.226%		$A =$	0.9051		
σ	25.0%				$XN(-d_2) - FN(-d_1)$	0.006306
T	1		Notional	100,000,000	Receiver swaption	570,713
X	5.417%				(in bps)	57.1

Figure 6.11: The Par Rate in a 2nc1 when $\sigma = 25\%$

flexibility of the swaps market and the creativity of those who structured trades. Index amortizing swaps were heavily used by certain regional banks as a synthetic investment vehicle, a substitute for a variety of mortgage bonds typically held by these financial institutions.¹⁷ It has been suggested by some

¹⁷For a further discussion of the use of index amortizing swaps, see, for example, "Banc One Corporation," Harvard Business School Case Study, February 24, 1994.

that the use of index amortizing swaps raised concern about how some financial institutions were able to use such option-embedded swaps without them being marked-to-market under then-current accounting guidelines.¹⁸ Broad changes in the accounting of derivatives were subsequently instituted under FAS 133. Index amortizing swaps were notoriously difficult to value, and pricing observed among competing dealers was sometimes surprisingly disparate.¹⁹

6.3.1 An Explanation of the Trade

An *index amortizing swap* (often referred to as an *index am* for short) is a swap whose notional can decline over time in accordance with a schedule. The schedule is driven by the level of a *reference rate* of interest, such as 3-month LIBOR, as observed on particular reset dates. Although the schedule is agreed to at the time the trade is executed, the actual amortization that occurs over the life of the trade depends on the path taken by the reference rate of interest. There may be a *lockout period* during which no amortization can occur.²⁰

The easiest way to see how an index am works is to go through a particular example. Consider \$100mm of an index am that has a final maturity of five years and a lockout period of one year. There can be no amortization of the notional during the first year of the swap. Assume that the reference rate is 3-month LIBOR and the *base rate* is 5.25%. After one year and on each subsequent coupon date, we will look at the level of 3-month LIBOR to determine the percentage of the remaining notional that amortizes in accordance with the schedule shown in Table 6.5.²¹ Linear interpolation of

¹⁸See, for example “Untangling the Derivatives Mess,” *Fortune Magazine*, March 20, 1995.

¹⁹Perhaps it would be more appropriate to describe the pricing from different dealers as shockingly different. In the swaps market, if a client asks one or more dealers to compete on a particular trade, it is common etiquette for the client to provide “color” as to where the trade got done to the losing dealer(s) who competed. At times, the variation in levels between the winning dealer and other dealers was so large (15, sometimes 20 bps or more in terms of the fixed coupon quoted) that some losing dealers could not reconcile the pricing under any parameterizations of their models. In fact, it was believed by some market participants at the time that the levels printed by at least one winning dealer sometimes broke arbitrage bounds governing the magnitude of the fixed coupon that could rightly be paid in such trades. Such a phenomenon suggests that the models employed by such a dealer to price such trades were flawed.

²⁰An index amortizing swap should not be confused with an *amortizing swap*, the latter of which has a declining notional balance set in accordance with a predetermined schedule. In an amortizing swap, there is no uncertainty regarding the declining notional of the swap. In an index amortizing swap, the declining notional of the swap is uncertain, depending on the path of the underlying reference rate over the life of the swap.

²¹This sample schedule is relatively simplistic. More complicated schedules were often used in actual trades.

Level of 3-month LIBOR	Percentage of Remaining Notional that Amortizes
$\leq$ Base rate – 100 bps	100%
Base rate	50%
$\geq$ Base rate + 100 bps	0%

Table 6.5: Index Amortizing Swap Schedule

this schedule is used to determine amortization for levels of 3-month LIBOR that do not explicitly appear in this table. For example, if LIBOR were to set at 4.75% on one of the reset dates after the lockout period, then 75% of the remaining notional would amortize. In an index am, the investor typically receives fixed and pays floating to a dealer.

Now let's look at what this schedule implies for an investor who is receiving fixed in an index am. When the market rallies, this typically benefits an investor who is receiving fixed in a swap. However, as the market rallies, the notional balance may experience significant amortization according to the schedule, which hurts the investor.²² That is, although the investor may be receiving a high fixed coupon in a low rate environment, the outstanding notional upon which he receives this fixed rate is diminishing.

On the other hand, when the market sells off, this typically hurts an investor who is receiving fixed in a swap. However, when the market sells off, the schedule indicates that the outstanding notional balance is going to be higher than the forwards originally predicted when the trade was first structured, which hurts the fixed rate receiver.

There are parallels, then, between receiving fixed in an index am and owning a mortgage security. Both exhibit this type of phenomenon, which is referred to as *negative convexity*, in which the trade shortens up in a rally precisely when we wish it would not, and the trade lengthens in a sell-off precisely when we wish it would not.²³ The investor who receives fixed in an

²²The implicit assumption being made here is that when the market rallies, all rates (including 3-month LIBOR) decline. However, even if the spot level of 3-month LIBOR does not decline as the market rallies, the forward LIBOR rates will decline. Models commonly employed to price index amortizing swaps will in such circumstances predict that the notional in subsequent periods will decline based on the lower forwards.

²³There is an important but subtle difference between the negative convexity implicit in this trade and the short option position embedded in other trades such as a cancelable swap. In a cancelable swap, one counterparty has sold an option to the other, and the option will be exercised (and the trade thereby canceled) at the discretion of the counter-

index am is compensated for enduring this negative convexity by receiving a higher coupon than he would in a plain vanilla swap.

The index amortizing swap was the swap market's answer to the mortgage market in the 1990s. That is, the index am is a trade that was designed to mimic some of the characteristics found in many mortgage securities (i.e., negative convexity) and provide a high yield (i.e., coupon) to the investor.²⁴ Some investors who had typically invested in mortgages sometimes found more attractive opportunities to receive fixed in index ams, depending on the relative value of options in the mortgage market and the swaps market.²⁵

6.3.2 Pricing Index Amortizing Swaps

Pricing index amortizing swaps is extremely complicated and beyond the scope of this discussion.²⁶ The declining notional feature of the trade from one period to the next based off of the remaining notional implies path dependency. As a testament to this complexity, the pricing shown by various dealers who priced index ams in the mid-1990s often varied significantly.

Index ams are typically not modeled analytically. If we were to envision pricing an index am in a binomial tree framework (in which nodes in the tree represent possible interest rates observed over the life of the trade), the setup would be pretty complicated. At each node we would have to keep track of, among other things, the path of 3-month LIBOR (for purposes of both computing amortization according to the schedule as well as determining payment on the floating side of the swap). We might consider parameterizing the magnitude of moves in our tree based on the level of implied volatilities trading elsewhere in the swaps market. This would be an important input used to determine the possible paths that LIBOR rates and swap rates could take over the life of the trade. We could use this sort of methodology to

party who owns the option. In an index am, no one really owns an option; the outstanding notional will decline not based on an option being exercised by a counterparty but, rather, based on what happens in the market.

²⁴Index ams have some different features as compared to mortgages. For one, amortization in mortgage securities is driven by the level of prepayments, which are relatively difficult to predict. In an index am, amortization is driven by a single reference rate, typically 3-month LIBOR. While it may very well not be easy to predict 3-month LIBOR in the future, some investors liked the fact that the cash flows in an index am could be fully specified as a function of LIBOR alone.

²⁵To put it another way, the relative attractiveness of index ams versus mortgages depended on the level of implied volatility trading in the swaps market and the mortgage market. In general, as swaption vol increased versus mortgage vol, this would tend to make index amortizing swaps a relatively more attractive alternative.

²⁶For a more detailed discussion of the pricing of index amortizing swaps, see, for example, Longstaff and Schwartz (2001), Fernald (1993–1994), Galaif (1993–1994), and Flavell (2010).

determine the fixed coupon for a given reference schedule that gives the trade an $NPV = 0$.

6.3.3 Relationship Between Index Amortizing Swaps and Cancelable Swaps

It can be shown that for index ams and cancelable swaps with the same final maturity and same lockout period (during which no amortization can occur in the case of an index am and during which the swap cannot be canceled in the case of a cancelable swap), the fixed rate in the cancelable swap must be higher than the fixed rate in the index am. We will suggest some intuition for the result. A cancelable swap is “more callable” than an index am. In a cancelable swap, the remaining notional on the trade can go from 100% to 0% at a single moment in time and typically does so when it is disadvantageous for the fixed rate receiver. In an index am, although the remaining notional on the swap may disappear as quickly, it is typically the case that the amortization is less drastic at any one time, since partial amortization of the swap can occur. Both trades tend to amortize when it is against the best interests of the fixed rate receiver, but the cancelable swap experiences a more binary, “all or none” amortization, whereas the index am can experience partial amortization.

Figure 6.12 shows the relationship between the fixed coupon obtainable in an index am as a function of a change in the base rate versus the coupon in a comparable cancelable swap. Both trades have a final maturity of three years and can't be called in their first year. The level of the 1-year and 3-year swap rates are also depicted. As we look to the right of this figure, we are considering cases in which the base rate is set very high. Take a look back at Table 6.5 and think about what this means for the index am. With the base rate set very high, it is a virtual certainty that the index am will completely amortize at the first opportunity to do so. That is, the index am would be expected to amortize completely at the end of one year. Note that in this range the coupon that would be paid in an index am is very close to that of the 1-year swap rate. At the other extreme, look to the left of this figure, in which we consider cases in which the base rate is set very low. With a very low base rate, it will be very difficult for the index am to experience any amortization whatsoever. In this case, we would expect the index am to remain a bullet swap with a maturity of three years. Note that the coupon obtainable by the index am in this range is very close to the coupon in a plain vanilla 3-year swap.

It is in the middle region when the index am becomes more interesting. When the base rate is set at more moderate levels, the amortization of the

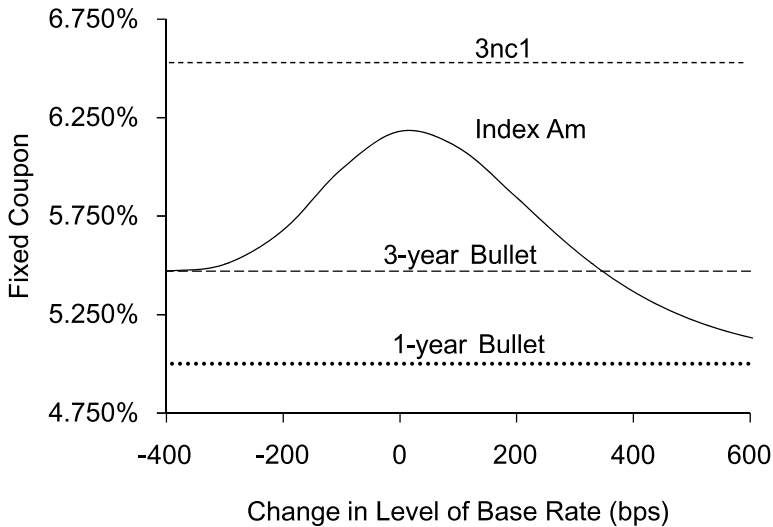


Figure 6.12: Coupon Comparison Between Index Amortizing Swaps, Cancellable Swaps, and Bullet Swaps as a Function of the Base Rate in the Index Amortizing Swap

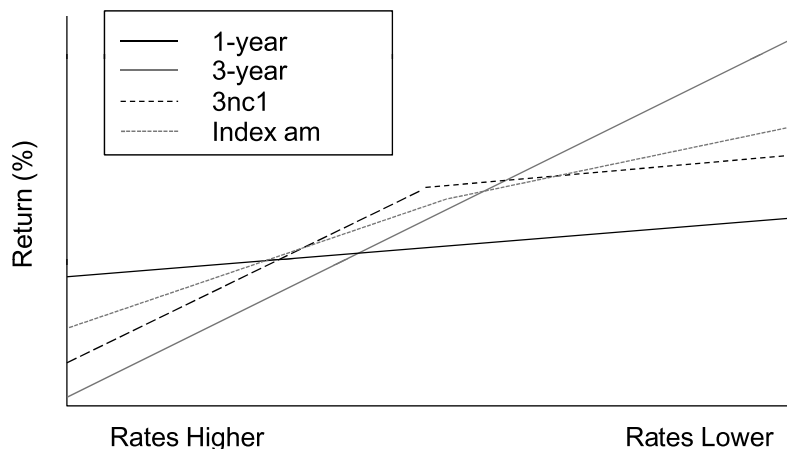
notional becomes less certain, and the value of the negative convexity that is shorted by the fixed rate receiver increases. It is in this region that the fixed rate in an index am can start exhibiting a significant pickup above both the 1-year and 3-year swap rates. Part of the job of someone structuring such a trade is to find a schedule that will pay a high fixed rate so as to be attractive to investors. Note that in all cases the coupon obtainable in an index am is always less than the coupon in the cancelable swap.²⁷

Return Comparison

Figure 6.13 shows a return profile comparison between 1-year swaps, 3-year swaps, a 3nc1, and a comparable index am with a 3-year final maturity and a 1-year lockout.²⁸ The 1-year and 3-year bullet swaps are represented by the

²⁷As alluded to earlier, it is believed by some in the industry that there were times in the 1990s during which at least one dealer would sometimes mistakenly show fixed coupons in index ams that broke this boundary condition. In particular, they sometimes showed a fixed rate in the index am that was higher than the fixed coupon in a comparable Bermudan cancelable. This was suggestive of a major problem with one's index am model.

²⁸We haven't discussed the concept of return for swaps, but for our purposes here we can think of it as any net coupon interest earned till the horizon date plus the mark-to-market of the swap at that time, divided by the notional of the swap.



Parallel Shift in Swap Rates

Figure 6.13: Return Comparison Between Bullet Swaps, Cancelable Swap, and Index Amortizing Swap. Figure Not Drawn to Scale.

straight solid lines. The 3-year has a higher slope than does the 1-year, due to the fact that the longer-dated swap has a higher PV01 and is therefore more sensitive to changes in interest rates. The 3nc1 cancelable swap is represented by the dark kinked dashed line. In a low rate environment, the return profile of the 3nc1 is parallel to but greater than that of the 1-year swap, and it is lower than that of the 3-year swap. In a high rate environment, the return profile of the 3nc1 is parallel to but greater than that of the 3-year swap but less than that of the 1-year swap. The intuition for this is as follows: as rates fall, it is likely that the 3nc1 will be called away and therefore will behave like a 1-year swap. However, the coupon of the 3nc1 is greater than that of the 1-year swap, and therefore the return profile of the 3nc1 is above that of the 1-year swap. In a rally when we are receiving fixed, we wish that we were in a swap that will remain outstanding and therefore have a high PV01, and that is why the return of the 3-year swap is superior in this case.

Conversely, as the market sells off, it is likely that the 3nc1 will not be called away and therefore will behave like a 3-year swap. However, the coupon of the 3nc1 is greater than that of the 3-year swap, and therefore the return profile of the 3nc1 is above the 3-year swap in this region. In a sell-off when we are receiving fixed, we wish that we were in low PV01 swap, and that is why the return of the 1-year swap is superior in this case.

In summary, when we are receiving fixed in the 3nc1, as the market rallies our trade will act like a 1-year swap even though we wish we were in a 3-year swap, and as the market sells off, our trade will act like a 3-year swap even though we wish we were in a 1-year swap. It is only in a rather tight range in which rates do not move much that the 3nc1 outperforms both the 1-year and 3-year swap consistently. The index am return profile is shown as the lighter kinked dashed line. Its return profile is less severe than the cancelable.

6.4 Knockout Swaps

A *knockout swap* is a fixed-floating swap in which payments on each side can be “knocked out” (meaning, they will not be made) if a reference rate reaches a certain level. There are several varieties of knockout swaps, including *period-by-period knockout swaps* in which payments may be knocked out one period but do not affect that status of subsequent payments and *full knockout swaps* in which all subsequent payments in a swap are knocked out once a reference rate has reached a certain level. It’s a trade whose heyday is surely behind us but that we see occasionally today.

When a client and dealer enter into a knockout swap, it is typically the client that is receiving fixed. Usually the knockout level of the reference rate is set “low,” in terms of the spot levels of rates but, more important, in terms of pricing from the forwards. In this case, the swap will knock out when the market rallies. In a standard fixed-floating swap, as the market rallies, the value of the swap increases to the fixed rate recipient. But it is precisely in these instances when the payments in a knockout swap disappear. In order to compensate the fixed rate recipient for this knockout feature, the fixed rate in a knockout swap is greater than that in a comparable maturity standard fixed-floating swap. The fixed rate recipient in a knockout swap is presumably entering into the trade because he does not believe that the reference rate will ever set low enough such that payments will be knocked out and is willing to take this risk in exchange for receiving the high fixed coupon.

Example

Let’s work through a concrete example of a period-by-period knockout swap. Recall our stylized example, assume that the swap curve is as given in Table 6.1, and further assume that the base vol curve is flat at 20%. Consider a \$100mm 4-year swap in which a client receives fixed annual bond and pays 1-year LIBOR to a dealer. This swap has the following knockout feature:

if, on any LIBOR reset date, 1-year LIBOR is less than or equal to the *knockout trigger* of 4.75%, then neither the fixed nor the floating payment is made for that period.²⁹ If payments are knocked out in any one period, they can resume in any subsequent period and will, provided that LIBOR is above 4.75%. To summarize, the structure of the swap is as shown in Table 6.6.

Figure 6.14 shows the net coupon each year from the point of view of the dealer as a function of where LIBOR sets.³⁰ The net cash flow at the end of the first year based off of today's LIBOR setting, denoted $L(0, 1)$, is known with certainty and is³¹

$$\text{Notional} \times [L(0, 1) - \text{Fixed}].$$

For the remaining years, the position depicted in Figure 6.14 is equivalent to a package consisting of³²

1. Paying fixed in a plain vanilla swap and receiving LIBOR
2. A long position in a floor struck at 4.75%
3. A long position in a binary floor struck at 4.75% that pays out an amount equal to the difference between the fixed coupon and the knockout trigger of 4.75%. That is, this option pays out the quantity

$$\text{Fixed} - 4.75\%$$

when LIBOR sets below 4.75% and zero otherwise, where Fixed is the fixed coupon that we are solving for in this trade.

To verify that these three components equate to the payoff diagram in Figure 6.14, consider the payoff of these three components at maturity, as shown

²⁹To avoid any confusion, if, for example, LIBOR sets below 4.75% one year from now, then there will be no cash flows in the swap two years from now.

³⁰When we draw Figure 6.14, how do we know the fixed rate we will solve for is greater than 4.75%? Assume that the fixed rate is less than or equal to 4.75%. Suppose the dealer receives on a 1-year swap at the par rate of 5% and pays on the knockout swap. At the end of the first year, he receives 5% in the 1-year swap and pays the fixed rate in the knockout swap (the first LIBOR set is 5% and thus the first period does not knock out), which is net positive cash flow. In subsequent years, if LIBOR sets above 4.75%, the dealer will be paying the fixed rate and receiving LIBOR, which is positive cash flow. When LIBOR sets below 4.75%, the swap knocks out, which is zero cash flow. This would be an arbitrage.

³¹Today's LIBOR setting is 5% and is thus above the knockout trigger of 4.75%, and therefore the cash flow at the end of the first year does not knock out.

³²See Appendix A for a review of pricing a trade as a package, or combination of options and cash.